TEAM NEXORA · DOSM DATATHON 2026 · FINAL ROUND

Judge Q&A Preparation

30 predicted questions with short answers, grouped by topic. The Q&A is 5 minutes after the 10-minute presentation, so expect about 4–6 questions. Answers use only facts from our report, results and dashboard.

> **How to use this.** Every member reads all 30 answers, not only their own area. Each question shows the marking-scheme codes it affects (PJT = presentation, DEM = demo, ANL = analysis, IMP = impact, INV = creativity). ★ **important** marks the 13 questions most likely to be asked or most likely to cost marks; learn these first. Square brackets [ ] mark things a teammate must confirm before the day.

> **Answering rules.**
> - **Two or three sentences, then stop.** Let the judge ask more.
> - **Say "goes with" or "is associated with", never "causes".**
> - **Own the limits.** "We haven't tested that yet; we would check it by…" scores better than guessing.
> - **Use the numbers:** +0.08 · 16 / 3 / 1 · 15,969 (about 16 thousand) · under 12,000 at 1–2 months · 16.8% · 20 countries · 2,300 country-months · 4-month delay (3 tested).
> - **Point to a backup slide** (13 Accuracy, 14 Markets, 15 Data & design, 16 Demo screenshots) when a number helps.
> - **Share the answers:** one person answers, a teammate may add one sentence only if it adds something.

Confirm before the day

Item	Ask
How the 20 countries were chosen	Data teammate: ranked by what, over which years, and why exactly 20? (The official data shows our 20 are ranks 1–20 by total arrivals from 2020 to May 2026.)
Any hyperparameter tuning	Model teammate: were the settings tuned, and on what data?
Odd dashboard labels and axes	Dashboard teammate: which ones, and the honest explanation.
Who did what	All: one sentence each.

1 · Problem, goals and scope

Owner: whoever wrote the report introduction. Backup slide: none needed.

1 What problem are you solving, and why does it matter? PJT2 · IMP2 ★ important

Tourism is 15.9% of Malaysia's economy and 3.7 million jobs. One national arrivals number can hide what is happening in individual countries, so planners can't tell whether a fall is broad or from a few markets, and can't see it early. We test the risk link country by country and forecast 1–4 months ahead.

2 Why only 20 countries, and how were they chosen? ANL1 ★ important

They are the 20 largest source markets by total arrivals in Tourism Malaysia's official data, together about 95% of arrivals in the official data from 2020 to May 2026. The same method can extend to more countries. Our totals are for these 20 only, not all of Malaysia. [Confirm the selection period with the data teammate before the day.]

2 · Data

Owner: data teammate. Backup slide 15 (Data & design).

3 Where does your data come from? Is it official?

ANL1 · IMP3

Arrivals: Tourism Malaysia with the Immigration Department. Exchange rates: Bank Negara Malaysia. Risk index: Caldara and Iacoviello. Oil: World Bank. Petrol and diesel: Ministry of Finance. GDP and jobs: DOSM's Tourism Satellite Account. All open and public. The source of every variable is recorded in a 63-entry catalogue.

4 How did you handle missing values?

ANL2 · ANL4

There were very few. Our 20 countries have a published arrival figure for every month from March 2017 to May 2026. The only blanks are June–September 2026, which are the months we forecast, so they were left empty, not filled. Five zeros during COVID are real border-closure months, kept as zeros. Exchange-rate gaps were filled from official observations (Bangladesh uses a documented cross-rate), and the original data is kept alongside.

3 · Method and model choices

Owner: model teammate. Backup slides 13 (Accuracy) and 15 (Data & design).

5 How do you stop the model from seeing the future?

ANL2 · ANL4 ★ important

Every input is used only after its real publishing date. Tourist arrivals are delayed by 4 months, risk and oil by 1 month, and training rows are used only if their target month was already published at the simulated date. Many projects skip this and look more accurate in testing than they could be in real life.

6 Why use two models, Ridge and LightGBM?

ANL2 · ANL4 ★ important

Each does one job. LightGBM gives the lowest forecast error but is made of trees, so it has no single number for the effect of risk. Ridge is a straight-line formula, so each country gets one readable risk slope. Both use the same inputs and the same data.

7 Why not LSTM, deep learning, ARIMA or Prophet?

ANL2

We have about 2,300 country-months, which is small for deep learning. Tree models trained on many short series together have a strong record for this kind of data. ARIMA fits one series at a time and can't share patterns between countries. We compared against simple rules and Ridge, as the literature recommends. We didn't test deep learning or ARIMA; that would be a reasonable extension.

8 Why one shared model for all countries instead of one per country?

ANL2 · ANL4

Each country has only about 111 usable months, too few to learn well alone. Sharing lets small countries borrow patterns from large ones, and the country input still lets each one behave differently.

9 Did you tune the hyperparameters? Isn't there a risk of overfitting?

ANL2 · PJT5

The settings are fixed and modest: 300 trees with a maximum depth of 4 and a learning rate of 0.05, with the same settings in every test. Overfitting is checked by testing forward in time on dates the model had not seen. [Confirm with the model teammate whether any tuning was done, and say so honestly.]

4 · Findings: forecasting accuracy

Owner: model teammate. Backup slide 13 (Accuracy).

10 How accurate is your model? Is 16.8% good?

ANL3 · PJT4 ★ important

Across the five normal-period tests, the forecast is off by about 16 thousand tourists per country per month, which is 16.8% of total arrivals. It is about 7% better than just repeating last month, and most reliable 1–2 months ahead (under 12,000). It is a modest improvement, not a breakthrough, and we say so.

11 Is LightGBM really better than Ridge?

ANL3 · PJT5

It had the lowest overall error (15,969 vs 16,727) and the lowest RMSE. But it is a fair win, not a big one: no model wins every horizon, the gaps at 2–4 months are under 2%, and Ridge wins 7 of the 20 individual tests. We chose LightGBM because one model makes all four forecasts, so the lowest total error matters most.

12 Why does the error double at 3–4 months?

ANL3

The further ahead, the less the latest data tells us, and more can change. At 1–2 months the model leans on recent arrivals; at 3–4 months that information is older. That is why we say it is most reliable 1–2 months ahead.

13 Why did repeating last month beat your model during COVID?

ANL5 · PJT5 ★ important

COVID was a sudden border closure that never appeared in the past data, so no model could learn it. Repeating last month reacts immediately to what just happened. The lesson is that in a sudden shock the latest month is the safer guide, so a forecasting tool should show reliability for each horizon. We report this openly.

14 Your forecast has no range. Isn't a single number misleading?

ANL5 · IMP3

Yes, it is a limitation. Forecasts are single best estimates. We label them as such, show the test errors by horizon so users see how far to trust them (about 12,000 per country at 1–2 months), and forecast ranges are our first next step.

15 What does your model forecast, and does it make sense?

ANL3

About 2.28 million arrivals from our 20 markets in June, 2.29 million in July, 2.17 million in August and 2.05 million in September 2026, stable then easing. This matches the usual seasonal pattern. Singapore is about 44% of the June forecast.

5 · Findings: the risk result

Owner: model teammate. Backup slide 14 (Markets).

16 What does your main finding mean?

ANL3 · PJT4 ★ important

Taken together, risk and arrivals show almost no link (+0.08). Country by country, holding oil, exchange rate, season and past arrivals constant, 16 countries send fewer tourists when risk is higher, 3 (Singapore, Indonesia, China) send more, and Thailand is about zero. The drops and rises cancel out in the total.

17 Is the risk effect statistically significant? ANL3 · PJT5 ★ important

Ridge doesn't give p-values, and we haven't calculated confidence ranges yet. The direction is the same for 16 countries, but some effects are small: India is close to zero, as is Thailand. The next step is a bootstrap that resamples whole months, because risk is one number for all countries each month.

18 Does risk cause the changes? ANL5 · PJT5 ★ important

No. These are associations. Risk moves together with the economy, oil and the post-COVID recovery, and we did not use a design that can prove cause. We say "goes with", not "causes".

19 Why do Singapore, Indonesia and China go up when risk goes up? ANL5 · PJT5 ★ important

We don't claim risk causes it. Possible reasons, which we have not tested: they are nearby, familiar markets with many short trips; some travellers may switch to nearby Malaysia when distant places feel risky; and their recovery after COVID, including visa-free entry for China from December 2023, happened in the same years risk was rising. We'd test this by refitting on recent years only.

20 The global risk index is the same for every country. How can each country have its own effect? ANL4

We multiply risk by each country in the model, so each country gets its own slope, learned from how its arrivals moved with risk over time. Because it comes from timing, a country's own trend could line up with risk by chance, which is another reason we call these associations.

21 If risk is only the sixth most important input, does it matter? ANL5

For the size of arrivals, past arrivals and season matter more, which makes sense. The risk result is about direction by country: it differs between markets and cancels out in the total. That is what the national figure hides.

22 What is SHAP and what does Nepal's 68% mean? ANL4

SHAP splits each prediction into each input's contribution. For Nepal, the model's average risk contribution is about 825 tourists, but Nepal's average arrivals are only about 1,200, so the ratio looks huge. It shows the model's use of risk relative to size, not real-world sensitivity, and ratios for tiny markets are noisy.

6 · Dashboard and demo

Owner: dashboard teammate. Backup slide 16 (Demo screenshots).

23 Why Power BI? What does the dashboard let users do? DEM2 · DEM5

Power BI works offline, with data built in, and standard visuals only. Four pages answer four questions: where visitors come from, what moves them, what is coming, and how far to trust it. Users can filter by region and country, click to cross-filter, and compare models and horizons.

24 Why do some charts show technical field names or odd axes? DEM4

[Dashboard teammate: know which labels and axes look odd and say honestly.] A safe answer: "Some labels still carry the raw field names and some axes need tidying. We'd clean these before real use." Only say it if true.

7 · Impact, SDGs and feasibility

Owner: report/impact teammate. Backup slide 15 (Data & design).

25 Who would actually use this, and how? IMP1 · IMP3 ★ important

Tourism Malaysia, to decide where and when to promote, by seeing which countries tend to send fewer tourists when risk rises. DOSM, to add a country-level view next to the national number. Hotels, airlines and homestays, to plan staff one to two months ahead, where the forecast is most accurate.

26 Have you talked to any of these users? IMP3 · PJT5

Not yet. These are possible uses based on the problem and the literature. We have not measured real impact, and we say that. Talking to users is the first step towards a pilot.

27 How would you measure the impact? IMP2 · IMP5

Our report lists indicators: tourism jobs, wages, skills and local business participation for SDG 8.9; energy and water per visitor, waste, emissions, crowding and resident views for SDG 12.b. A pilot would compare planning decisions with and without the forecast against these indicators.

28 How does this link to the SDGs? Isn't forecasting arrivals just promoting more tourists? IMP5 · ANL5

Not by itself. Higher forecast arrivals are a planning signal, not proof of sustainable progress. SDG 8.9: earlier planning supports steadier tourism jobs. SDG 12.b: the dashboard is a monitoring tool. The report says that real sustainability needs jobs, local benefit, emissions and carrying-capacity indicators added alongside.

8 · Creativity and AI

Owner: any member. Keep it short.

29 What is new or creative about your project? INV1 · INV2 ★ important

Two things: testing the risk link country by country, because the national total hides opposite movements, and a strict publication-date rule so the model never sees the future. Together they give honest, country-level forecasts, with the limits openly stated.

9 · Tough and team questions

Owner: whoever is closest to the topic. Stay calm, short and honest.

30 What is the biggest weakness of your project? PJT5 · ANL5 ★ important

Three: the risk results are associations, not causes; forecasts are single numbers without ranges; and we haven't yet tested whether the country effects are statistically significant or hold without the COVID years. All three are our next steps.